MLFF PERF-P0 Matched CPU Qualification

Complete supplied LTA TARGET-DATA2B workload

mdstats project

2026-08-15

1 Decision

PERF-P0 passes as bounded supplied-data TARGET-DATA2B authority.

- Source release: mdstats 0.20.179a0
- Target sources: **27**
- Target frames: **37,633**
- Target atoms: **6,322,344**
- Coverage families: **8**
- Family elements: **263,398**
- PERF-BASE0 numerical-array agreement: **exact, 48/48 arrays**
- Historical/P0 scientific-digest agreement: **exact**

This report is a human-readable projection of mlff_perf_p0_lta_cloud_cpu_2026-08-15.json. The JSON record is authoritative.

2 Measurement design

The complete supplied target corpus was ingested and cached once. Historical and PERF-P0 family construction then ran in isolated matched processes over the same in-memory scientific arrays. Each path has five retained runs.

The comparison separates:

1. scientific identity: array bytes, family content, and scientific digest; and
2. execution evidence: wall/process CPU time, RSS, worker/thread settings, and host state.

Execution evidence is excluded from the scientific digest. No favorable single-run timing is promoted as authority.

3 Scientific equivalence

Every historical/P0 run has scientific digest

2f04aba96b876a10e62b7b0c22f26b544836005d11bc893177d4b29cbe4a3f82.

All 48 numerical arrays match the PERF-BASE0 target_data2b_exact_radii stage byte-for-byte. These arrays cover frame indices, values, balanced weights, robust scales, exact local radii, and q01/q50/q99 extent statistics.

The PERF-P0 benchmark scientific digest is

ab5e57750a06a3895d6349846238073a7bdca40a1c5270409999bfd6507a1d05.

Execution digest:

4265881375205beb954222ac8d0b1372221c4779dcbc5567e9ff3a64d82681a1.

Whole-record digest:

`cd9240e5c4936e1ac22409df8d56b06f487debd78f150bc3a45f7423d0917db0`.

4 Exact family construction

Path	Median wall	Observed range	Median process CPU	Median peak RSS
Pre-P0 exact path	7.541 s	6.826–9.926 s	27.091 s	329.47 MiB
PERF-P0 exact path	6.236 s	5.818–8.253 s	26.043 s	328.50 MiB

The matched median wall improvement is

$$I_{wall} = 100\left(1 - \frac{t_{P0}}{t_{pre}}\right) = 100\left(1 - \frac{6.236470132}{7.541068095}\right) = 17.30\%.$$

The overlapping ranges show that operating-system scheduling and page-cache state remain material. The median result is therefore interpreted with the full five-run envelope, not as a deterministic speed ratio.

5 Persistence

The full reference was serialized and restored through both representations in isolated processes.

Representation	Write wall	Read wall	Bytes	Write RSS increment	Read RSS increment
Nested JSON v1	10.366 s	14.382 s	42,749,676	167.77 MiB	189.35 MiB
Native NPY v2	0.184 s	0.180 s	17,912,666	0.12 MiB	28.02 MiB

Derived effects:

Effect	Result
Write speed ratio	56.22x
Read speed ratio	79.70x
Serialized-size reduction	58.10%
Read RSS-increment reduction	85.20%

Both restored references have content digest

`4f46dfaa6c366ede1cd4d19cf5fb8be65cfe49067b5aff85d668e0078915f9b8`.

The exact v1/v2 migration report has no difference paths and digest

`bbe21f1c20beaefb7c837ec20330365853727366f07e4b8a795745aa048bfd88`.

6 Host boundary

The available cloud machine exposes an Intel Xeon Platinum 8573C, nine visible logical CPUs, an 8-core cgroup CPU quota, and a 4 GiB cgroup memory limit. Declared native thread pools were capped at eight threads. Python was 3.13.5; NumPy 2.3.5; SciPy 1.17.0; ASE 3.29.0.

These host and package observations are execution evidence. They do not enter TARGET-DATA2B scientific identity.

7 Limits

- The benchmark covers complete supplied-data label, cell, mobile/framework, and species-force families. Production DATA6 model-derived families require the authorizing checkpoint and complete campaign bundle and were not fabricated.
- Family construction is isolated from XML ingestion so historical and P0 paths receive identical arrays.
- The native read is authenticated and uses read-only mmap at the benchmark threshold.
- This gate does not qualify complete TARGET-DATA2C/DATA7 selection, DATA8, TRAIN2/EVAL2, GPU memory, utilization, OOM, or backoff evidence.

The normative method, invariants, and migration contract are defined in the PERF-P0 native target-coverage specification. `PERF-P1` is next.